# Monte Carlo Investigation of Criticality in the Two-Dimensional Ising Model

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This paper presents an investigation into the phase transition of the two-dimensional Ising model via Monte Carlo simulations. By leveraging both Metropolis and Wolff algorithms, we systematically analyze the evolution of magnetisation, spatial correlations, and thermodynamic observables near the critical point. Binder cumulant analysis yields an accurately determined critical temperature, $T_c = 2.269 \pm 0.001$, while log–log analyses produce critical exponents: $\beta = 0.117 \pm 0.001$, $\gamma = 1.744 \pm 0.020$, and $\nu = 1.01 \pm 0.01$. Although the magnetisation and specific heat exponents show slight deviations from the exact theoretical values ($\beta = 0.125$ and a logarithmic divergence in specific heat), these differences are ascribed to finite-size effects and limitations in fitting logarithmic behavior. Overall, the results underscore the efficiency of the Wolff algorithm in mitigating critical slowing down and validate the Monte Carlo method in calculating characteristic scaling laws of second-order phase transitions.

## INTRODUCTION

A phase transition is said to have occurred when a small change in a system's control parameter, such as temperature or pressure, results in a dramatic shift in the state of the system. For example, water transitions from a liquid to solid at 0°C, whereas in ferromagnets, cooling past a critical temperature causes the atoms' magnetic moments to align, resulting in spontaneous magnetisation. At these points of criticality, systems become highly sensitive to disturbances and exhibit recurring patterns across all length scales, where a microscopic snapshot reflects the same structure visible on a macroscopic scale.

The Ising model, introduced by Lenz and later studied by Ising [1], provides a mathematical framework for examining phase transitions. By reducing systems to fundamental local interactions, it reveals how large-scale phenomena emerge at critical points. A key insight from the Ising model is the concept of universality: despite differences in microscopic details, many systems exhibit the same behavior near their critical point. This reveals that phase transitions follow fundamental principles that extend across seemingly unrelated physical systems. By studying the Ising model, we can investigate these principles and gain a deeper understanding of critical phenomena. Given its historical significance, this paper will examine the Ising model in the context of ferromagnetism and present its analysis using the language of statistical mechanics.

## THEORY

The two-dimensional Ising model is defined on a simple square lattice, with each site taking a spin value of $\sigma_i = \pm 1$. These spins can be viewed as magnetic dipoles: where the total magnetisation M is simply the sum of individual moments,

$$M = \sum_i \sigma_i \quad . \tag{1}$$

At high temperatures, thermal fluctuations dominate and effectively randomize the spins, leading to zero net magnetisation. As the temperature is lowered, however, the tendency for spins to align (energetically favourable) begins to outweigh these randomizing effects, eventually leading to a spontaneous magnetisation below the critical temperature.

To explore how temperature affects the probability of different system states, we first define the system's total energy using the Hamiltonian,

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j \quad , \tag{2}$$

where the summation runs over pairs of nearest neighbors on the simple square lattice, and $J$ is a coupling constant. When $J > 0$, the system favors neighboring spins that are aligned, thereby promoting ferromagnetism at low temperatures. Using the defined Hamiltonian we can now construct the partition function:

$$Z = \sum_{\{\sigma\}} e^{-\beta \mathcal{H}} \quad , \tag{3}$$

where $\beta = 1/(k_B T)$, $k_B$ is the Boltzmann constant, and $T$ is the temperature. From the logarithm of the partition function, $\ln Z$, one can derive expressions for a system's thermodynamic observables. In particular, we focus on the magnetic susceptibility $\chi$ and specific heat capacity $C$:

$$\chi = \frac{1}{k_B T} \left( \langle M^2 \rangle - \langle M \rangle^2 \right), \tag{4}$$

$$C = \frac{1}{k_B T^2} \left( \langle \mathcal{H}^2 \rangle - \langle \mathcal{H} \rangle^2 \right). \tag{5}$$

The Ising model undergoes a second-order phase transition at the critical temperature $T_c$, where the magnetisation changes smoothly from zero to a finite value. Unlike first-order transitions, which feature abrupt changes in the order parameter, second-order transitions exhibit singularities in thermodynamic quantities such as the heat capacity and susceptibility. These quantities, which correspond to second-order statistical moments, develop sharp peaks or divergences at $T_c$, marking the transition.

To characterize how spin interactions extend over a distance, we define the two-point correlation function,

$$G(r) = \langle \sigma_0 \sigma_r \rangle \; - \; \langle \sigma_0 \rangle \langle \sigma_r \rangle. \tag{6}$$

This function indicates whether spins separated by a distance $r$ are likely to point in the same direction (positive correlation) or opposite directions (negative correlation), or whether they are effectively independent (zero correlation).

Near the critical point, these observables follow power-law behaviors characterized by critical exponents $\alpha$, $\beta$, $\gamma$ and $\nu$ (see below). Determining these exponents allows us to quantify how the system behaves near criticality, providing a concrete measure of the transition.

## METHODS

In a two-dimensional system composed of an $L \times L$ lattice of spins, where each spin can take one of two values, the total number of possible coniguration of the system is $2^{L^2}$. This exponential growth renders exact computation of the partition function infeasible for large $N$, motivating the use of Monte Carlo methods to effectively sample the state space, enabling accurate estimation of a system's thermodynamic properties.

One method well-suited to this problem is the Metropolis-Hastings Algorithm [2], a Markov Chain Monte Carlo (MCMC) Method. Starting from a randomly initialised state, each step involves proposing a new state by flipping a randomly chosen spin. The energy difference between the new and current state, $\Delta E$, is then computed. The proposed move is accepted unconditionally if $\Delta E \leq 0$, as it lowers the system's energy. Otherwise, it is accepted with probability:

$$P(i \to j) = e^{-\beta \Delta E}. \tag{7}$$

This acceptance rule allows the system to move to higher-energy states with some probability, ensuring proper exploration of the state space and preventing trapping in local minima.

The Metropolis algorithm satisfies detailed balance, ensuring that the Markov chain converges to the Boltzmann distribution, which describes the equilibrium state of the system. For any two states $i$ and $j$, this is expressed as

$$P(i)W(i \to j) = P(j)W(j \to i), \tag{8}$$

where $P(i)$ is the equilibrium probability of state $i$, and $W(i \to j)$ is the transition probability. Since spin flips are selected randomly, the proposal mechanism is symmetric (i.e., the probability of proposing a move from $i \to j$ is equal to $j \to i$). Thus, the acceptance probability alone enforces detailed balance.

At high temperatures ($\beta$ small), the acceptance probability is less sensitive to energy changes, leading to frequent transitions and broader sampling of the state space. At low temperatures ($\beta$ large), energy-increasing moves are rarely accepted, driving the system toward lower-energy states. Provided the Markov chain is ergodic and aperiodic [3], this process ensures convergence to the equilibrium Boltzmann distribution.

Near the critical temperature, the single-spin flip dynamics of the Metropolis algorithm suffer from critical slowing down, where changes to the system propagate slowly, requiring many updates to produce distinct configurations. To alleviate this, we employ the Wolff algorithm [4], which updates clusters of correlated spins simultaneously. In this method, a random seed spin is selected and its nearest neighbors are added to the cluster with a probability

$$p = 1 - e^{-2\beta J},$$

where $J$ is the coupling constant. The process is repeated iteratively until the cluster is fully grown, and then the entire cluster is flipped. As demonstrated by Wolff (1989) [4], the inherent symmetry of the cluster update process maintains detailed balance, ensuring convergence to the correct Boltzmann state.

To investigate the critical temperature, we introduce The Binder Cumulant $U_L$,

$$U_L = 1 - \frac{\langle M^4 \rangle}{3\langle M^2 \rangle^2} \quad . \tag{9}$$

This dimensionless quantity characterizes the probability distribution of the system, making it a useful tool for distinguishing between ordered and disordered phases [5]. At the critical temperature $T_c$, the system exhibits scale-invariant behavior, and the probability distribution of $M$ becomes independent of the system size. As a consequence, when $U_L$ is plotted as a function of temperature for different L, the curves intersect at $T_c$. This universal crossing point provides a robust and accurate estimate of $T_c$ that is less susceptible to finite-size effects and does not require prior knowledge of the critical exponents.

Once the critical temperature is determined, the behavior of key thermodynamic quantities near $T_c$ can be characterized by the critical exponents. The basic scaling relations near the critical temperature are:

$$M \sim (T_c - T)^{\beta} \quad \text{for } T < T_c, \tag{10}$$

$$\chi \sim |T - T_c|^{-\gamma}, \tag{11}$$

$$C \sim |T - T_c|^{-\alpha}. \tag{12}$$

For the correlation function, it typically follows an exponential decay near $T_c$:

$$G(r) \sim e^{-r/\xi}, \tag{13}$$

where $\xi$ is the correlation length. Near the critical temperature, $\xi$ follows the scaling relation:

$$\xi \sim |T - T_c|^{-\nu}. \tag{14}$$

By relating the behavior of observables near the critical point through these power laws, we can deduce the critical exponents that underpin the transition's universal dynamics.

## RESULTS

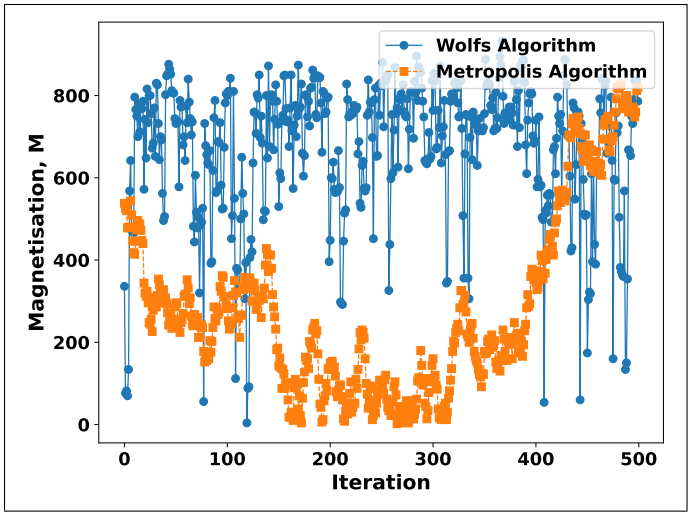



FIG. 1. Evolution of magnetisation versus Monte Carlo steps for the Wolff and Metropolis-Hastings algorithms at T = 2.75, L=32.

Figure 1 shows the comparison of the Wolff algorithm to the Metropolis-Hastings algorithm in tracking magnetisation evolution over 500 iterations. The plot illustrates that the Wolff algorithm's cluster update method reaches equilibrium rapidly, while the Metropolis-Hastings algorithm's individual spin update converges more slowly and displays greater fluctuations before stabilizing. This contrast underscores the Wolff algorithm's efficiency in overcoming critical slowing down, making it particularly advantageous for determining the critical temperature under time constraints.

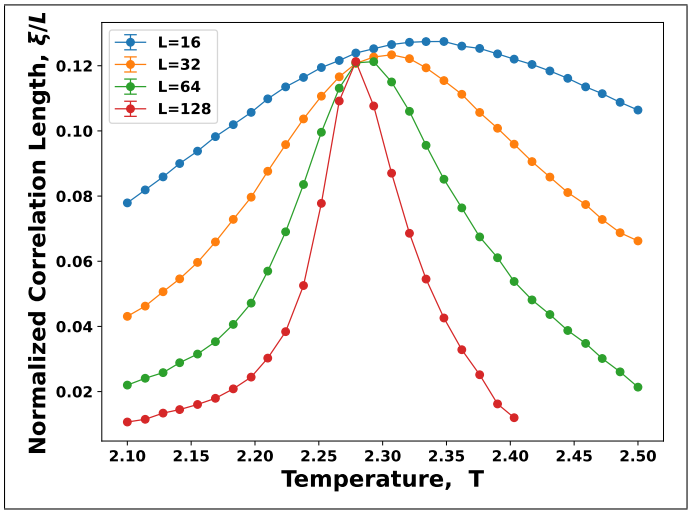



FIG. 2. Extracted (Normalised) correlation length $\xi$ as a function of temperature, for system sizes L = [16, 32, 64, 128].

At both high/low temperatures, the connected correlation $G(r)$ decays rapidly (FIG.2), confirming the dominance of short-range order. Near $T_c$ = 2.269 however, the decay slows on all length scales, indicating the onset of long-range correlations.

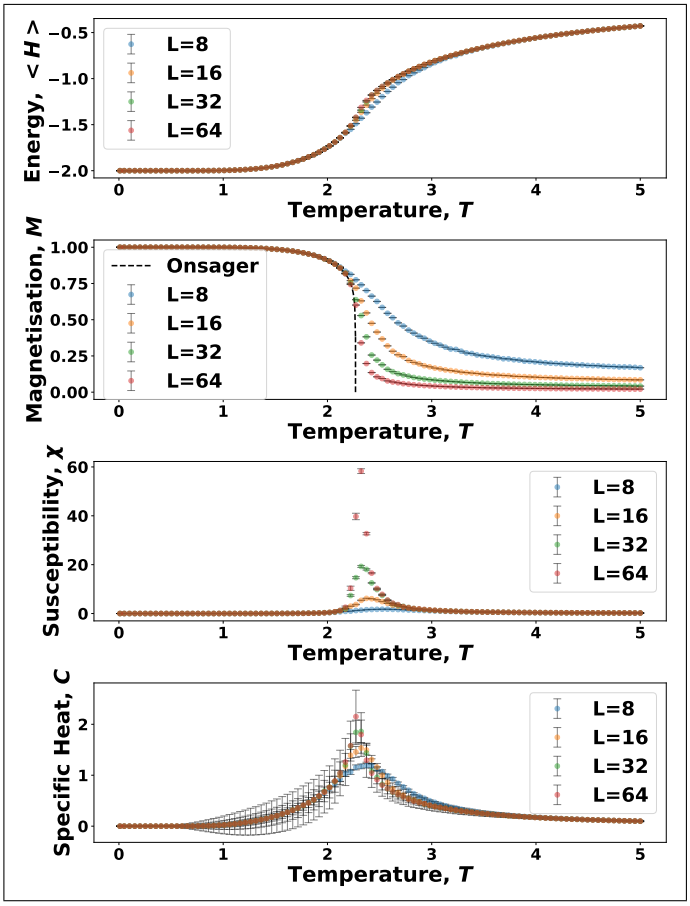



FIG. 3. Temperature dependence of magnetisation, susceptibility, specific heat, and internal energy, for system sizes L = [8,16,32,64]. Lars Onsager's analytical solution [6]

The susceptibility and specific heat display pronounced peaks at $T_c$ (FIG.3), reflecting the divergent fluctuations expected at a second-order phase transition. The magnetisation curve exhibits a continuous onset of spontaneous order below $T_c$ and follows Onsager's analytical solution below $T_c$, while the internal energy also shows smooth behavior with no discontinuities.

Plotting the Binder cumulant as a function of temperature for various system sizes reveals the expected invariance in the probability distribution at the critical point (FIG.4). Analysis of the intersections yields an average critical temperature of $T_c = 2.269 \pm 0.001$ in close agreement with Onsager's analytical prediction of 2.269 [6]. The slopes obtained from the linear fits in the log-log plots yield estimates for the critical exponents $\beta$, $\gamma$, and $\nu$ (FIG.5). The correlation length exponent is determined to be $\nu = 1.01 \pm 0.01$, in accurate agreement with the theoretical prediction of $\nu = 1$. The magnetisation exponent is estimated as $\beta = 0.117 \pm 0.001$, which is slightly lower than the exact value of 0.125, likely due to finite-size effects and the limited range of reduced temperatures used in the fit. The susceptibility exponent is found to be $\gamma = 1.744 \pm 0.020$, matching closely with the expected

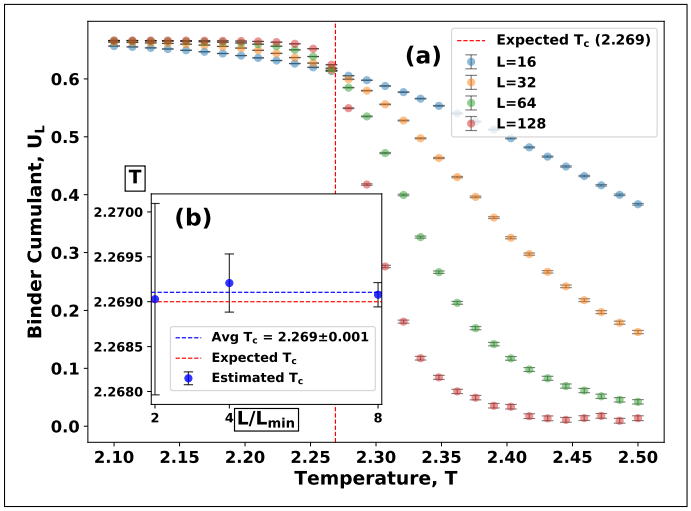



FIG. 4. (a) Binder cumulant $U_L$ versus temperature, for system sizes L = [16, 32, 64, 128].(b) Intersection Temperatures for different system sizes with $L_{min} = 16$

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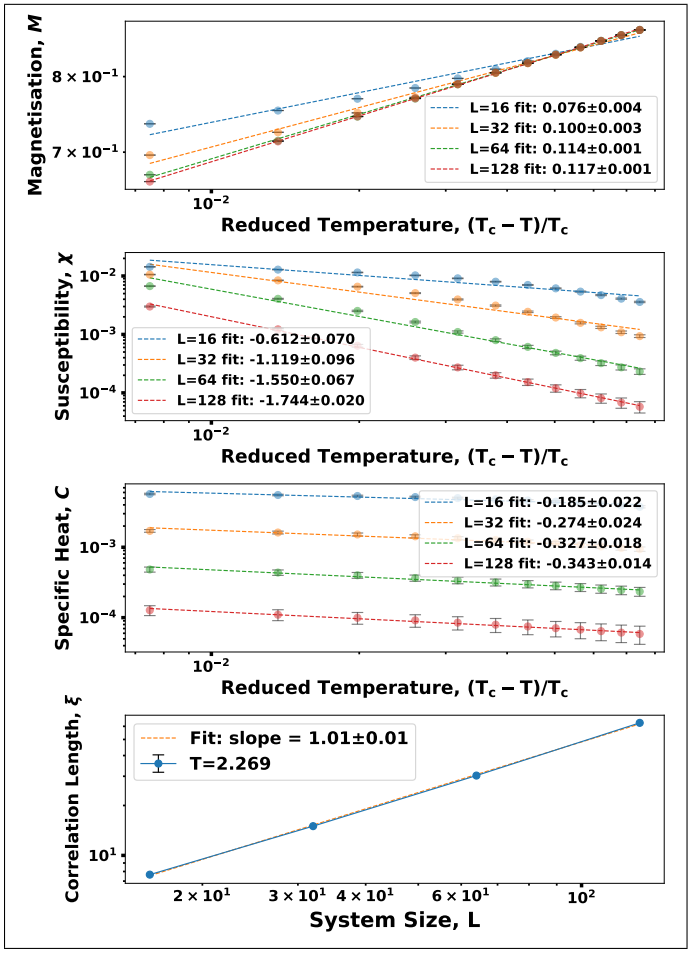



FIG. 5. Log–log plots for magnetisation, susceptibility, and correlation length used to extract the critical exponents $\beta$, $\gamma$, and $\nu$.

value of 1.75. In contrast, the specific heat displays an effective exponent of approximately $\alpha \approx 0.343 \pm 0.014$, deviating from the theoretical expectation of a logarithmic divergence ($\alpha \approx 0$). This discrepancy can be attributed to the challenges of fitting a logarithmic divergence using a power-law approximation over a finite temperature range.

## DISCUSSION

The overall results demonstrate that the Wolff algorithm efficiently overcomes the critical slowing down observed with the Metropolis method near $T_c$. Spatial correlation analyses confirm the divergence of the correlation length and the onset of long-range order near the critical point. Binder cumulant analysis robustly identifies the critical temperature, and the log–log analysis yields critical exponents consistent with theoretical predictions.

The close agreement between our estimates for $\nu$ and $\gamma$ with theoretical predictions showcases the accuracy of the MCMC method. The slight underestimation of $\beta$ and deviations in the specific heat exponent highlight finite-size effects and fitting limitations. Alternative approaches, such as multicanonical sampling or improved estimators [7], could reduce biases and enhance precision near the critical point.

## CONCLUSION

Our study of the Ising model via Monte Carlo simulations elucidates the behavior of ferromagnetic phase transitions. By leveraging the Metropolis and Wolff algorithms alongside Binder cumulant analysis, we construct a robust framework for understanding critical phenomena. These methods not only capture key scaling behaviors but also highlight the strengths and limitations of different sampling techniques in studying phase transitions.


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